

AYUSHI SINGH

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EDUCATION

Manipal Institute Of Technology, Karnataka

October 2021 - September 2025

Bachelor of Technology in Electronics and Communication Engineering

EXPERIENCE

Rockwell Automation

May 2024 – July 2024

Industrial Trainee

Noida, Uttar Pradesh

- Programmed and optimized an automated bottle-filling system in Studio 5000 on ControlLogix PLCs, achieving 4 bottles per 15s and automating truck-loading in 5s with real-time monitoring and logic tweaks.
- Built, tested, and debugged a 4-way traffic light controller in Studio 5000, strengthening hardware logic design and real-time control skills.
- Supported the 435 KTPA Smelter Expansion by reviewing GA drawings and executing cold/hot testing (continuity, power-up, I/O checkout, BOOTP/network configuration) to validate panel layouts, signalling, and safety compliance.

PROJECTS

AdCanvas AI | Repo | *React, JavaScript, CSS, Gemini REST API, Git*

November 2025 - Present

- Built a React multi-step campaign setup UI with CSS styling to capture structured inputs and generate ad copy variants via prompt templates.
- Integrated REST API calls to the Gemini API; handled request/response formatting, JSON parsing, and error states for stable generation.
- Delivered a working proof of concept through iterative fixes; improved reliability by handling edge cases (empty inputs, retries, failed responses).

Predictive Maintenance Using Machine Learning | Link

April 2025 - September 2025

- Built an end-to-end predictive maintenance classifier on AI4I-2020 CNC sensor data with Type engineered from Product ID, identifiers removed, and a stratified split.
- Handled heavy class imbalance using **SMOTE** only inside training folds which prevented data leakage.
- Trained Random Forest, HistGradientBoosting, and XGBoost; selected the operating threshold from the precision-recall curve; test PR-AUC ≈ 0.97 and ROC-AUC ≈ 0.99 , with recall ≈ 0.94 at precision ≈ 0.98 .
- Explained the model with SHAP; ToolWear, Torque, and RotationalSpeed were the main drivers while temperatures were minor, and we summarized maintenance actions from these insights.

Customer Churn Analysis | Link

August 2024 – September 2024

- Built interactive Power BI churn dashboards using Power Query and DAX to track churn KPIs and segment-level trends.
- Developed a logistic regression churn model with 85% accuracy, generating risk scores to prioritize retention outreach.

SKILLS

Technical:	Python, C++, JavaScript, SQL, OOP, DSA, DBMS, Computer Networks, REST APIs
Tools:	Git/GitHub, VS Code, Jupyter, Excel, Tableau, Postman
Testing/Dev:	Unit Testing (PyTest), Debugging, Linux, Deployment (Railway)
Non-Technical:	Problem Solving, Client Communication
Certifications:	McKinsey Forward Program; Machine Learning with Python (IBM)
Competitions:	GSSoC '24 - Open-source contributor (merged PRs). Myntra HackerRamp - Proposed Gen-Z personalization and sustainability concept